Politics, Education, and Health: Factors Associated with COVID-19 Vaccination

AN EXPLORATORY ANALYSIS USING COUNTY-LEVEL CDC AND CENSUS DATA

CHRISTOPHER HENSON, M.S.
ROBINSON LANDING BUILDING ENGINEER

Author's Note

This report was independently researched to examine factors associated with COVID-19 vaccination rates at the county level in the United States. Data sources include the CDC, the U.S. Census Bureau's American Community Survey, the U.S. Department of Agriculture's Rural-Urban Continuum Codes, and the MIT Election Data and Science Lab. Analysis was performed using R and related packages. I want to thank these organizations for making their data and tools publicly available.

The goal of this report is to analyze patterns in vaccination uptake. I encourage incorporating critical thinking, media literacy, and statistical reasoning into education to support future generations in making evidence-based decisions.

- Chris Henson, M.S., Statistical Science, George Mason University '23

Summary

In March 2020, the World Health Organization (WHO) declared COVID-19 a pandemic. In the years that followed, millions of people worldwide would lose their lives while every aspect of daily life would be disrupted. The speed at which safe and effective vaccines against the virus were developed and distributed was an incredible achievement of science, public health, and international cooperation. However, in the United States, vaccination rates varied wildly across the country. This project aims to identify potential reasons behind this, with the goal of better informing responses to possible future health crises. Using 2023 county-level data, this analysis indicates that political affiliation was the strongest predictor of vaccination rates. The counties with higher support for President Donald Trump in the 2020 Presidential election tended to have lower vaccination rates, even when accounting for other factors such as race, educational attainment, median income, age, and other health factors.

Data & Methodology

County-level COVID-19 vaccination rates and the prevalence of health factors, such as obesity, were obtained from the Centers for Disease Control and Prevention (CDC). Demographic and socioeconomic data, including racial composition, educational attainment, and income, were gathered from the United States Census Bureau's American Community Survey (ACS) 5-year estimates for 2023. Election data were sourced from the MIT Election Data and Science Lab (MEDSL), which provided county-level results for the 2020 election. Lastly, the United States Department of Agriculture (USDA) supplied data on the rurality of each county with the Rural-Urban Continuum Code (RUCC). The CDC stopped tracking COVID-19 vaccination in May 2023; therefore, nearly all data used for this project are from 2023 to maintain temporal consistency. However, in June 2022, the United States Census Bureau recognized Connecticut's councils of governments as county equivalents for statistical purposes. In contrast, the CDC continued to group its data by the traditional county. This was addressed by using the ACS 5-year estimate from 2021 for Connecticut only.

The response variable for all analyses in this project was the percentage of the county population fully vaccinated against COVID-19. A person is considered fully vaccinated if they have received the complete first series, which typically consists of two doses for most versions of the vaccine, but only one dose for some. Important predictor variables included political affiliation, measured by the percentage of votes for Donald Trump in the 2020 election; educational attainment, measured by the percentage of those 25 and older with a bachelor's degree or higher; and the age-adjusted prevalence of selected health factors among adults, such as obesity and diabetes. Demographic variables include racial composition (percentage

identifying as non-Hispanic White, Black or African American, Hispanic, Asian, and others) and age distribution, including a derived variable that measures the percentage of individuals aged 75 and older. Rurality was identified through the USDA's RUCC code, a discrete scale ranging from the most urban (rated as 1) to the most rural (rated as 9) counties. All predictor variables used in this analysis were selected based on previous research and their apparent relevance.

Multiple linear regression modeling was used to analyze the relationship between COVID-19 vaccination and the predictor variables. The variance inflation factor (VIF) was used to assess multicollinearity among the independent variables, while the adjusted R-squared was used to measure the model's explanatory power. Eighteen of the more than 3,000 counties in the United States did not have COVID-19 vaccination data and were removed from this analysis. The remaining counties were stratified into two groups based on the percentage of votes for Donald Trump: "pro-Trump" (55% or more) and "non-Trump" (< 55%). This analysis aimed to determine which factors influence vaccination rates while controlling for political affiliation.

Analysis

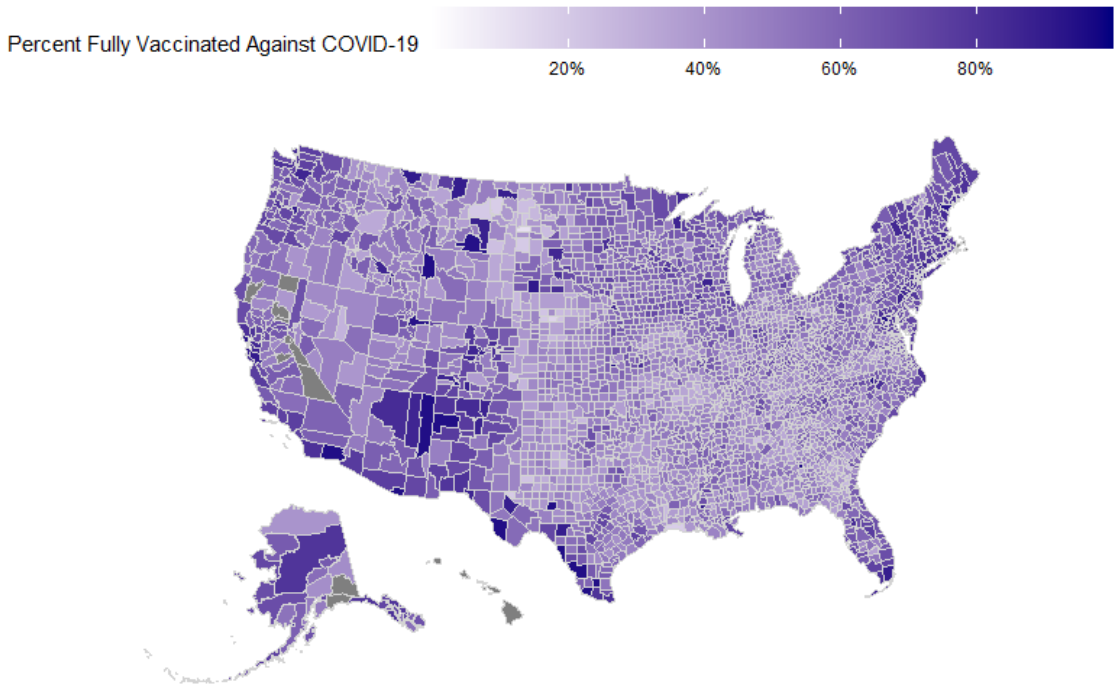

Figure 1: COVID-19 Vaccination Rate by County as of May 2023, grey indicates no data available

As shown in Figure 1, the COVID-19 vaccination rate varied widely across the United States. Counties in the Northeast, Southwest, and on the West Coast generally had higher vaccination uptake, while those in the Deep South and the rural Midwest had the lowest. This pattern

highlights the importance of analyzing vaccination rates at the county level, rather than the state or national level, as significant variations exist within states and even between neighboring counties.

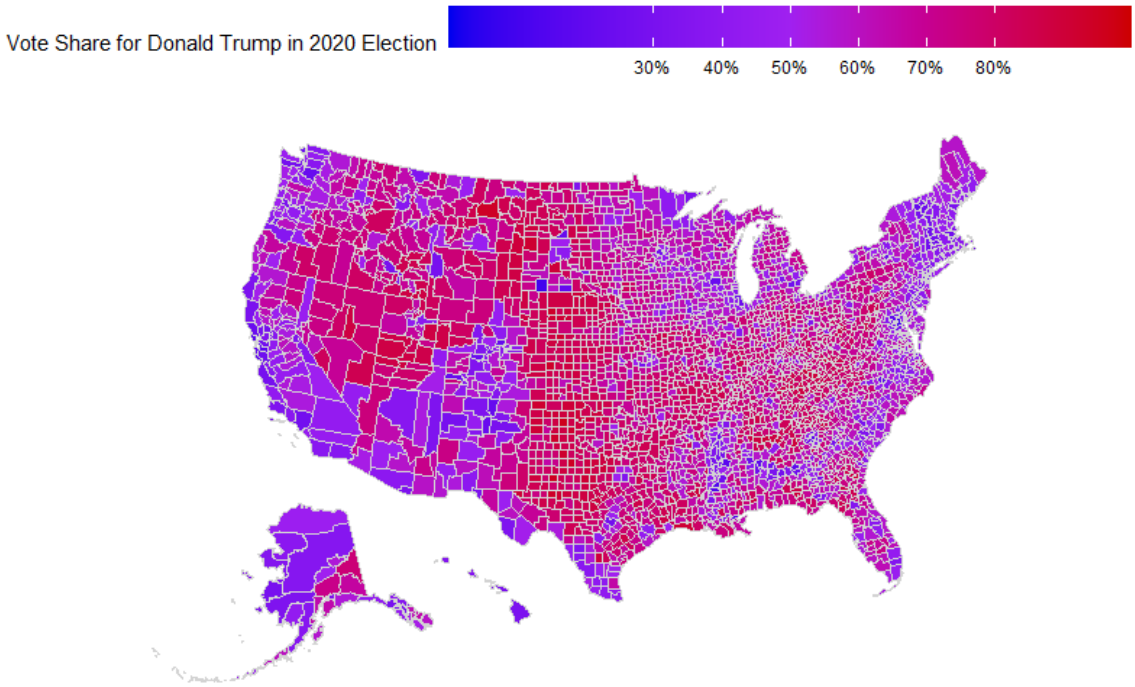

Figure 2: Map of 2020 county-level presidential election results by Trump vote share

Figure 2 illustrates the county-level election results of the 2020 presidential election. Note that large areas in northwest Texas, Kansas, and Nebraska strongly supported Donald Trump in 2020, while Figure 1 shows that these regions have among the lowest vaccination rates in the country. In contrast, counties along the coast of California, northern New Mexico, and much of New England had higher-than-average vaccination rates and strongly favored Joe Biden in 2020. Comparison of Figures 1 and 2 suggests a relationship between political affiliation and COVID-19 vaccination uptake, which can be explored more directly through a scatterplot.

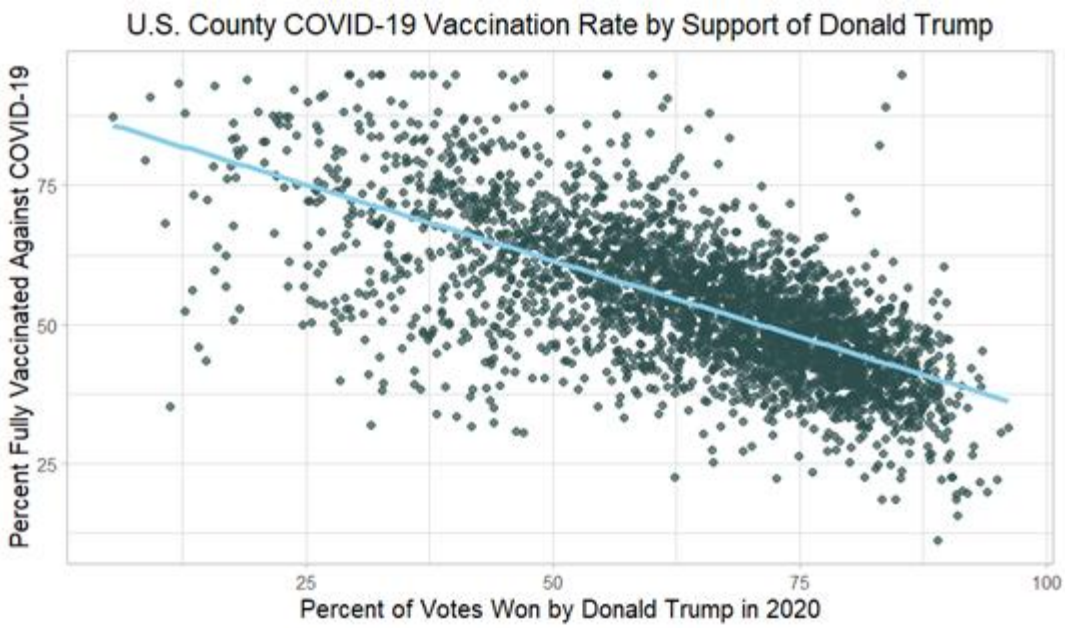

Figure 3: Scatterplot comparing Trump vote share with vaccination rate

Figure 3 illustrates a clear negative correlation between support for Donald Trump in the 2020 election and the county-level COVID-19 vaccination rate. Notice how the points cluster more tightly when Mr. Trump's vote share exceeds 55%, suggesting a stronger and more consistent relationship in these counties. A simple linear regression model quantifies this relationship: for every 10 percentage-point increase in Trump's vote share, there is approximately a 5.5 percentage-point decrease in the COVID-19 vaccination rate.

Table 1: Simple linear regression of vaccination rate and Trump's vote share

Intercept	**Coefficient**	**p-value**	**Adjusted R-Squared**
88.8155	-0.54645	$p < 0.0001$	0.4609

This simple model accounts for approximately 46% of the variation of COVID-19 vaccination rates between counties, suggesting that political affiliation alone is strongly associated with vaccine uptake while leaving space for other possible factors. A multiple linear regression model was constructed to investigate the potential impact of specific demographic, economic, and health factors.

Table 2: Multiple regression for all counties with political, educational, and health factors

Variable	**Coefficient/Estimate**	**p-value**	**VIF**
Intercept	95.456	$p < 0.0001$	n/a
Percent for Trump	-0.453	$p < 0.0001$	1.41
Percent with bachelor's or above	0.183	$p < 0.0001$	2.36
Age-adjusted obesity rate	-0.451	$p < 0.0001$	1.83

The variables used in this model, in addition to political affiliation, are educational attainment (percentage of county residents aged 25 or older with a bachelor's degree or higher) and the age-adjusted obesity rate among adults living in the county. Other demographic and health factors were considered but not used due to their minimal contribution to explanatory power. This model accounts for approximately 52% of the variation in county-level COVID-19 vaccination rates, a modest but meaningful improvement over the simple model. The variance inflation factor (VIF) was used to check for any overlap between the predictor variables. Although there is some correlation, the highest VIF is 2.36, which is well below the values commonly cited as problematic.

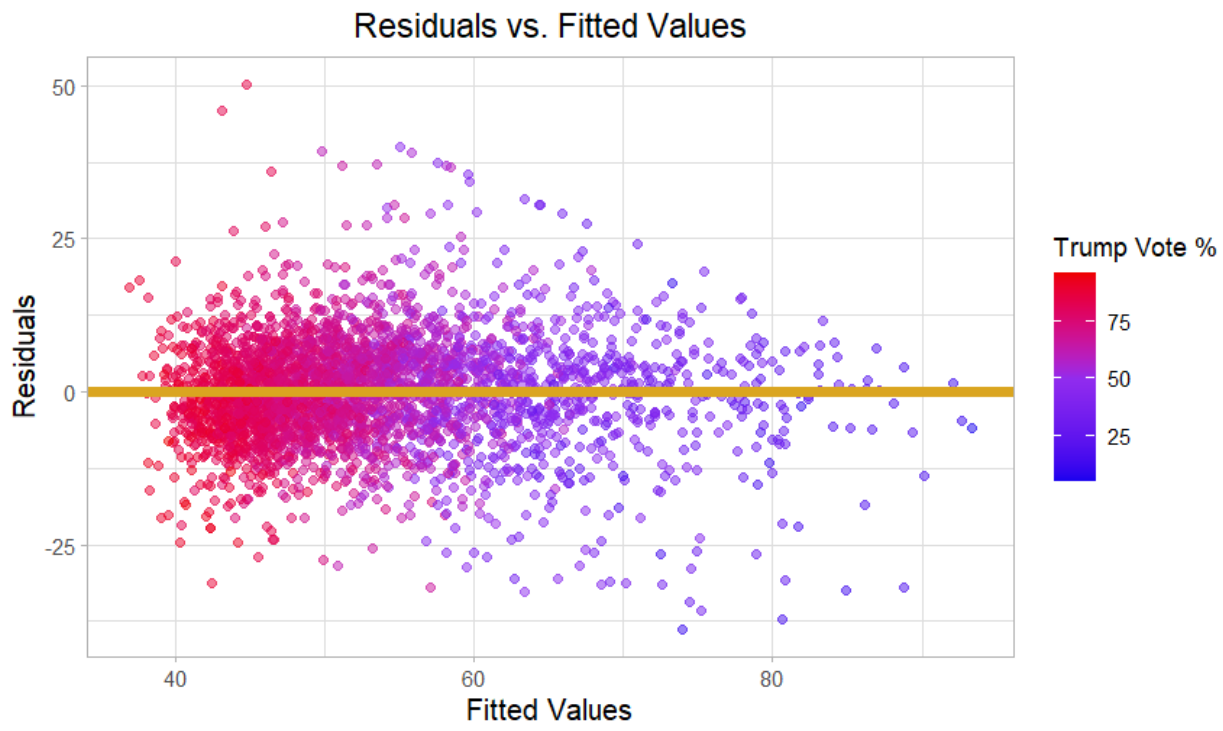

Figure 4: Fitted Values vs. Residuals, colored by Trump vote share

Figure 4, which plots residuals against fitted values, shows that while the model captures the overall variation in COVID-19 vaccination rates, counties that did not support Mr. Trump exhibited more variability and did not fit the model as neatly as those counties that strongly supported him. This suggests heteroskedasticity, meaning the model's variance may differ across political contexts. An interesting observation is that many jurisdictions with significantly lower vaccination rates than predicted are actually "independent cities" in Virginia, which are not part of any county. One possible reason for this discrepancy is that people were vaccinated in neighboring counties or were misclassified by data collectors, despite the CDC's goal to count individuals based on their place of residence. This highlights the potential limitations of

geographically based analysis, particularly in unique jurisdictions such as independent cities. However, it also suggests that grouping counties by political affiliation may yield more consistent results than simply grouping geographically.

To further investigate this pattern, the data were stratified by Mr. Trump's vote share. The median percentage of support for Donald Trump in the 2020 election was approximately 68%, which was primarily driven by the support of many sparsely populated counties (approximately 400) where he won at least 80% of the vote. The counties were first separated into two groups based on whether Mr. Trump's vote share was at least 55%. Those below 55% will be referred to as "non-Trump" counties, while those with 55% or more will be considered "pro-Trump" counties for this analysis. The 55% threshold was chosen because counties below this level demonstrated greater variance in their vaccination rates, as illustrated by Figure 3. Stratifying the data in this way highlights whether the relationship between COVID-19 vaccination and political affiliation is substantially different between counties that strongly supported Donald Trump and those that did not.

When the counties were stratified, a simple linear regression model revealed a weak relationship between Trump support and COVID-19 vaccination rates in the "non-Trump" counties (adjusted R-squared = 0.032). In contrast, the same model for "pro-Trump" counties showed a much stronger relationship (adjusted R-squared = 0.362). This suggests the broader national pattern is also reflected in counties that strongly supported Mr. Trump.

A multiple linear regression model was constructed to further analyze the "pro-Trump" counties. Political affiliation was retained, as its relationship with COVID-19 vaccination remains strong in these counties. Additional variables included educational attainment (as above), the poverty rate among senior citizens (aged 65 and above), and the age-adjusted percentage of the population without health insurance coverage. The adjusted R-squared for this model is 0.4115, indicating a modest improvement over the simpler version.

Table 3: Multiple regression for "pro-Trump" counties with political, educational, and socioeconomic factors

Variable	**Coefficient/Estimate**	**p-value**	**VIF**
Intercept	91.97	$p < 0.0001$	n/a
Percent for Trump	-0.601	$p < 0.0001$	1.13
Percent with bachelor's or above	0.223	$p < 0.0001$	1.37
Age-adj. percent without health coverage	-0.119	$p = 0.0003$	1.16
Poverty rate among seniors (65+)	-0.219	$p < 0.0001$	1.26

Among the predictors used here, only educational attainment had a positive association with COVID-19 vaccination rates. This implies that higher levels of education are associated with higher COVID-19 vaccine uptake in counties that favored Mr. Trump in the 2020 election.

Conversely, the higher poverty rates among senior citizens and fewer citizens with proper health insurance coverage are associated with lower vaccination rates, suggesting that economic and healthcare access barriers may have hindered vaccination efforts in these counties. However, even when accounting for socioeconomic and demographic factors, political affiliation remained the strongest predictor of vaccination rates. If all other variables remained equal, one would expect that for every 10-percentage-point increase in votes for Donald Trump, the expected COVID-19 vaccination rate fell by approximately six percentage points in the "pro-Trump" counties.

The counties where Donald Trump earned less than 55% of the vote exhibit different patterns in vaccination uptake. On its own, political affiliation does not appear to be a strong predictor of COVID-19 vaccination rates among "non-Trump" counties, as a simple model like the one used for the "pro-Trump" counties shows (adjusted R-squared = 0.076). However, when controlling for other demographic and socioeconomic factors, it emerges as a powerful predictor.

A multiple linear regression was then performed on the "non-Trump" counties. In addition to political affiliation, variables used in this model include the percentage of county residents who identify as Black or African American (non-Hispanic), the median household income in the county, and the percentage of county residents without health insurance coverage, adjusted for age. The resulting model shows strong predictive power, with an adjusted R-squared of 0.445.

Table 4: Multiple regression for "non-Trump" counties with political, demographic, and socioeconomic factors

Variable	**Coefficient/Estimate**	**p-value**	**VIF**
Intercept	75.761	$p < 0.0001$	n/a
Percent for Trump	-0.461	$p < 0.0001$	1.12
Percent identifying as Black (non-Hispanic)	-0.286	$p < 0.0001$	1.44
Median income in thousands of U.S. dollars	0.148	$p < 0.0001$	1.66
Age-adj. percentage without health coverage	0.265	$p < 0.0001$	1.3

The model results indicate that, when all other variables are constant, political affiliation had the strongest association with COVID-19 vaccination rates among the "non-Trump" counties. Specifically, vaccination rates were expected to decline by approximately 4.6 percentage points for every 10-percentage-point increase in Mr. Trump's vote share. Both the level of support for Mr. Trump and the percentage of residents identifying as Black or African American (non-Hispanic) were negatively associated with vaccine uptake. In contrast, median income and the age-adjusted rate of residents without health insurance coverage were positively associated with increased vaccination. The positive relationship between lack of health coverage and vaccination rate may reflect targeted outreach to lower-income areas of these counties.

Discussion

The analysis indicates that COVID-19 vaccination became deeply intertwined with political polarization in the United States. Across all U.S. counties, as well as within groups stratified by support for Donald Trump in the 2020 presidential election, his vote share consistently emerged as the most influential factor, with greater support associated with lower vaccine uptake. However, it is essential to note that this analysis does not attempt to determine causality. It cannot establish whether individuals who were already vaccine-hesitant were more likely to support Mr. Trump, or whether his supporters were influenced by political and social cues to forgo vaccination against COVID-19. However, the results reveal a strong association between political affiliation and vaccine choice, warranting further investigation. For example, surveys suggest that individuals who identify as Democrats were much more likely to receive a COVID-19 vaccine than Republicans.[1]

Public discussion about the COVID-19 pandemic and the government responses to it were highly polarized in the United States. Communication from political leaders, social media commentators, and traditional media pundits likely influenced the politically divergent attitudes towards potential preventative measures, including vaccination. For example, some polling has shown that the likelihood of an individual receiving a vaccine is correlated with their primary source for news about the pandemic.[2] It is beyond the scope of this analysis to assess the impact of any one person or statement. However, the strong association between political affiliation and COVID-19 vaccine uptake illustrates that social and political issues may shape individual health behaviors.

An issue that received greater attention during the COVID-19 pandemic was the spread of misinformation, primarily through online and social media platforms. A study published in *Scientific Reports* reveals that areas where users shared more misinformation and content from low-credibility sources had higher rates of vaccine hesitancy, and that this effect is observed in both majority Republican and majority Democratic areas.[3] Interestingly, the researchers observed that the relative impact of this was greater in regions with lower support for Donald Trump, possibly because the baseline rate for vaccine hesitancy was much higher in areas with greater support for him. This dynamic may help explain the positive relationship between

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[1] Cary Funk and John Gramlich, "10 Facts about Americans and Coronavirus Vaccines," Pew Research Center, September 20, 2021, https://www.pewresearch.org/short-reads/2021/09/20/10-facts-about-americans-and-coronavirus-vaccines/.

[2] Mark Jurkowitz and Amy Mitchell, "Americans who relied most on Trump for COVID-19 news among least likely to be vaccinated," Pew Research Center, September 23, 2021, https://www.pewresearch.org/short-reads/2021/09/23/americans-who-relied-most-on-trump-for-covid-19-news-among-least-likely-to-be-vaccinated/.

[3] Francesco Pierri et al., "Online Misinformation Is Linked to Early COVID-19 Vaccination Hesitancy and Refusal," Scientific Reports 12, no. 1 (April 26, 2022), https://doi.org/10.1038/s41598-022-10070-w.

educational attainment and vaccination rates, as individuals with higher education often have better access to reliable sources and are more equipped to evaluate conflicting information. Regardless, it is unfortunate that vaccine uptake appears to have been strongly influenced by factors unrelated to epidemiology and public health.

Beyond political affiliation and educational attainment, various demographic and socioeconomic factors were found to be associated with COVID-19 vaccination rates. In the model covering all counties, a negative correlation was found between the age-adjusted obesity rate among adults and the uptake of vaccines. This aligns with data indicating that "obese" individuals are less likely to participate in preventive healthcare, such as cancer screenings, due to real or perceived stigma against overweight people within the medical community.[4]

Within the "non-Trump" counties, the percentage of residents who identified as Black or African American (non-Hispanic) was also negatively related to vaccine uptake, as shown in Table 4. This reflects historical and systemic factors that have fostered greater mistrust of the government and medical institutions within the Black community, stemming from past unethical medical practices like the Tuskegee Syphilis Study.[5] Another potential reason for this is the increased barriers to quality healthcare, such as fewer primary care physicians in majority-Black communities compared to other areas.[6]

One notable finding of this analysis was that while the percentage of county residents without health insurance has a negative association with vaccination rates in "pro-Trump" counties, the relationship was positive in "non-Trump" counties. This may reflect stronger outreach or public health efforts in certain areas, though such factors are difficult to quantify at the county level. Another possible explanation relates to racial disparities in healthcare. According to the CDC, about 10.5% of Black individuals under 65 lack health insurance coverage, compared to just 6.5% of White individuals, meaning that Black individuals are about 60% more likely to be uninsured than non-Hispanic White individuals.[7] Concurrently, the median percentage of residents who identify as Black or African American (non-Hispanic) in "non-Trump" counties is approximately 17.6% while the median for "pro-Trump" counties is about 1.6%. Thus, when controlling for the percentage of Black or African American (non-Hispanic) residents, the percentage of residents without health coverage appears to have a positive association with

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[4] Robert D McGuigan and Jenny M Wilkinson, "Obesity and Healthcare Avoidance: A Systematic Review," AIMS Public Health 2, no. 1 (2015): 56–63, https://doi.org/10.3934/publichealth.2015.1.56.
[5] Carolyn M. Dolan and Sha'Meyah Bell, "Historical Trauma, Health Care Distrust, and the Legacy of Tuskegee: Implications for Nurse Practitioner Practice and Research," The Journal for Nurse Practitioners 20, no. 4 (March 9, 2024): 104910, https://doi.org/10.1016/j.nurpra.2023.104910.
[6] Darrell J. Gaskin et al., "Residential Segregation and the Availability of Primary Care Physicians," *Health Services Research* 47, no. 6 (April 23, 2012): 2353–76, https://doi.org/10.1111/j.1475-6773.2012.01417.x.
[7] Robin A. Cohen and Emily P. Terlizzi, "Demographic Variation in Health Insurance Coverage: United States, 2022" (Centers for Disease Control and Prevention, October 5, 2023), https://doi.org/10.15620/cdc:133321.

COVID-19 vaccination rates, as it primarily captures the uninsured populations among non-Black residents.

Support for Donald Trump in the 2020 presidential election consistently emerged as the strongest predictor of county-level vaccination rates, with higher support associated with lower vaccine uptake. Other components, including educational attainment, obesity, race, and socioeconomic factors, also contributed to this variation. These findings highlight the potential influence of political, social, and systemic divisions on the response to public health initiatives, even in high-leverage situations where clear scientific evidence is available.

This analysis is limited by its use of aggregate, county-level data, which cannot determine individual motivations or decision-making. Therefore, these findings should not be considered evidence of causation, but rather as an indication of broader societal patterns. Future research on this topic could build on this analysis by designing surveys on vaccination, political affiliation, media consumption, and other components to analyze individual behavior more thoroughly. Comparing the results of this analysis with state and local policies could reveal how these policies influenced COVID-19 vaccine uptake. Lastly, examining how trust in institutions has changed over time could offer insight into the current state of polarization in public health behaviors.

These results suggest how social and political identity, along with trust in institutions, may influence responses to public health crises more than solid scientific evidence. When public health issues become politicized, deciding to vaccinate can feel like an act of loyalty or identity affirmation rather than a choice based on science and safety. Protecting against disease has become less important than protecting one's sense of belonging. Trust in institutions has declined in recent years, often deservedly. However, much of the mistrust has been fueled by bad-faith actors. Investing in education in critical thinking, evidence-based reasoning, and media and scientific literacy from an early age could aid future generations in combating misinformation.

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